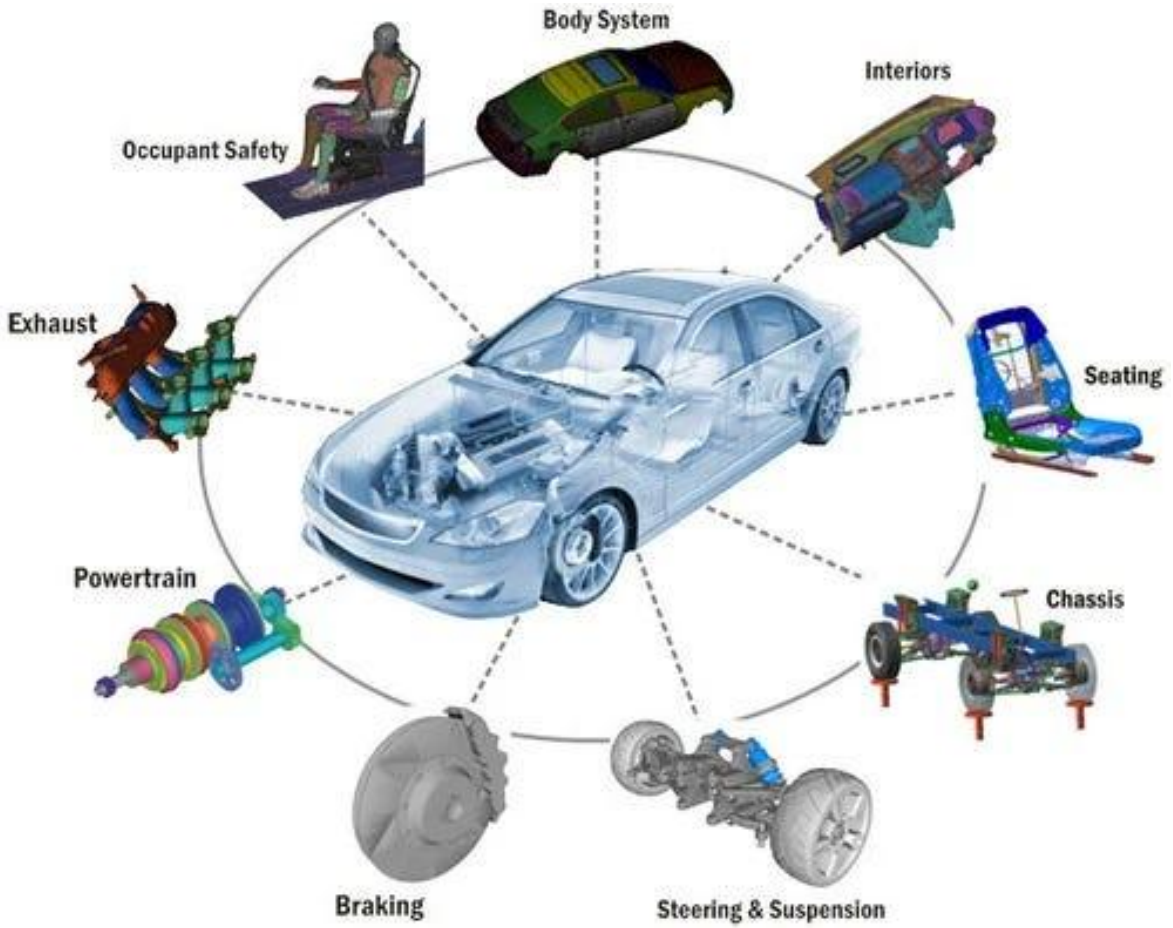
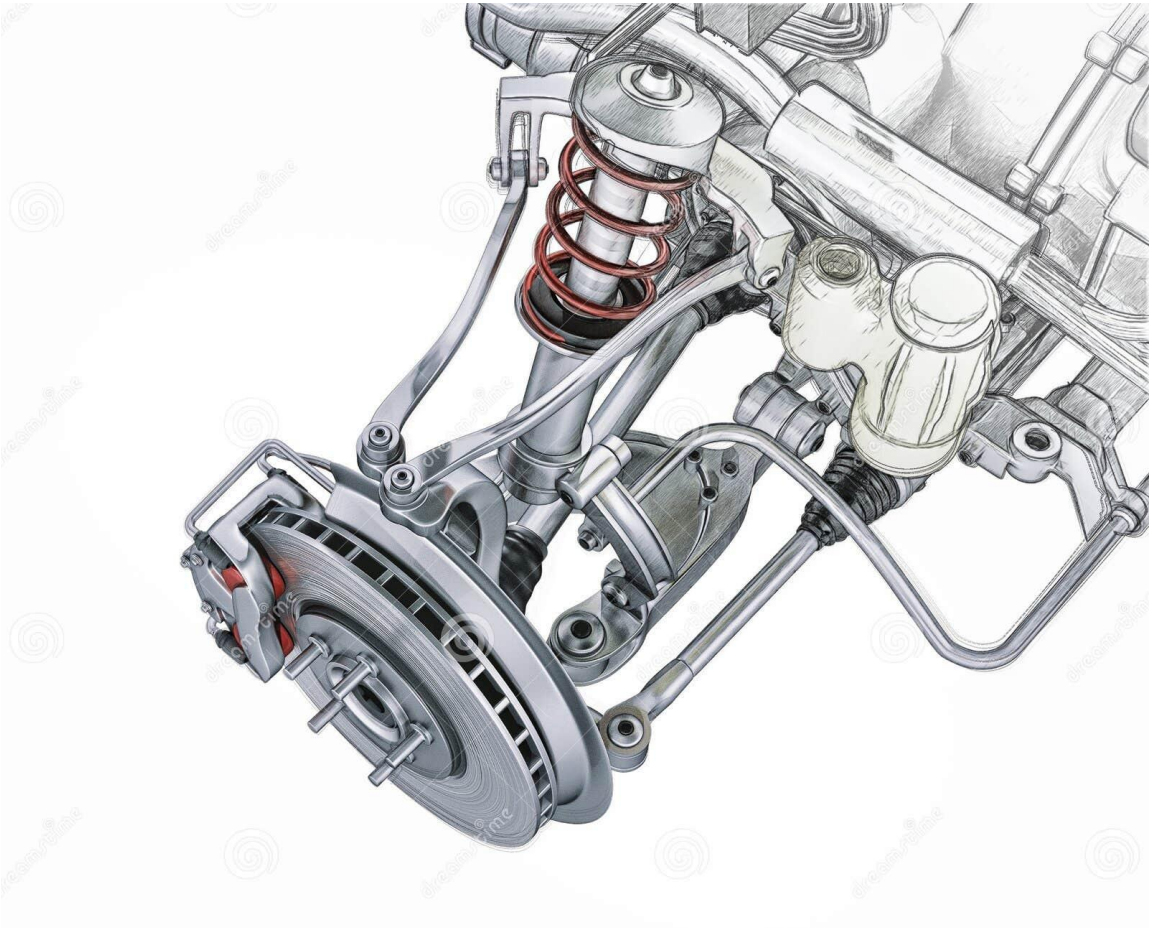




Automotive Engineering (MAM 336)

Week 2

Introduction to Transmission System

DR/ MOHAMED EL TAHER MOHAMED

Major Automobile Systems

Power plant / engine

Drive train: clutch, gearbox, shafts and differential

Electrical system: ignition, starting, lighting and battery

Wheels and tires

Suspension system

Steering system

Braking system

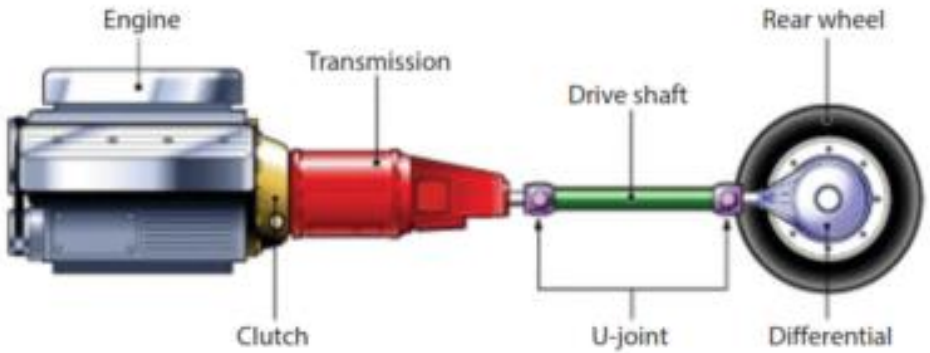
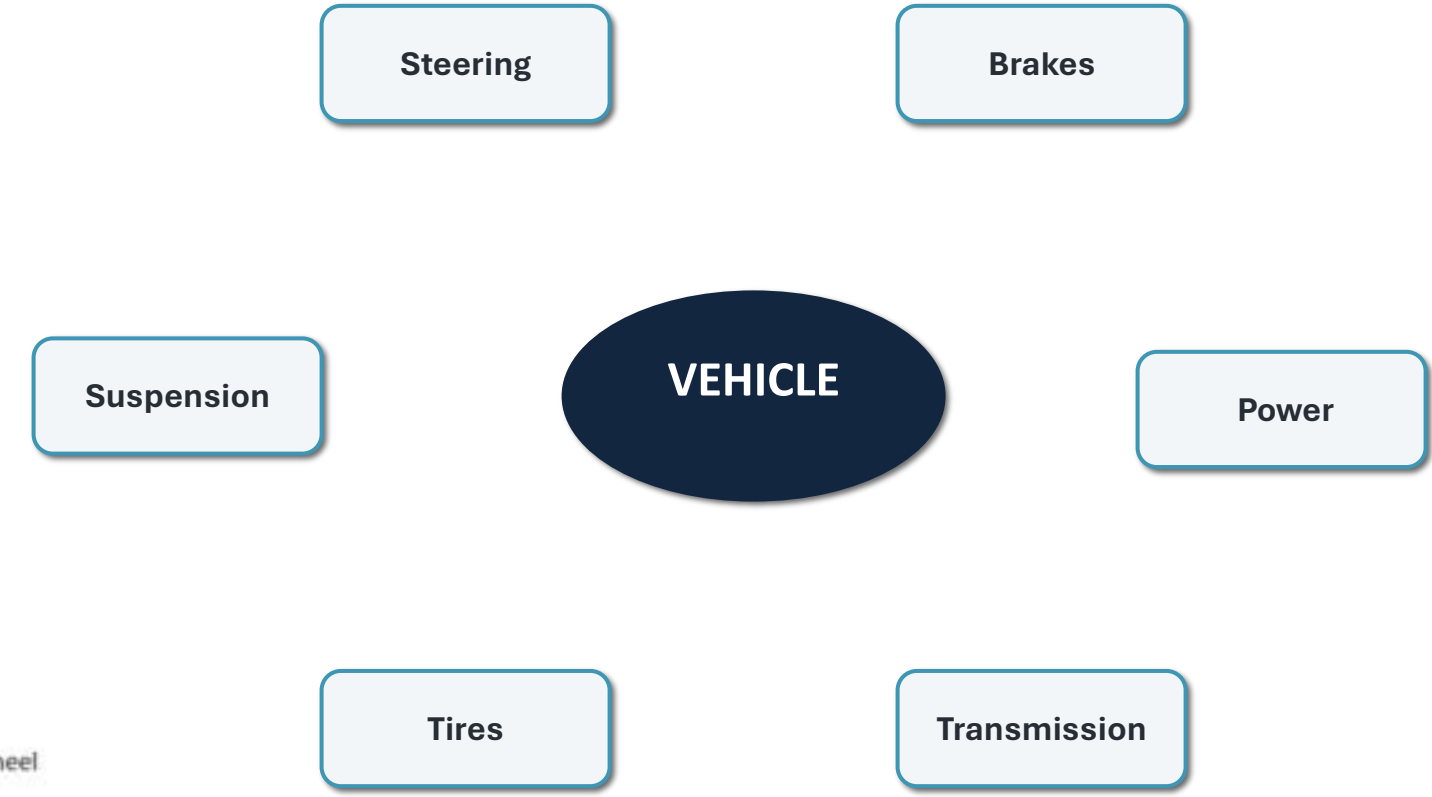
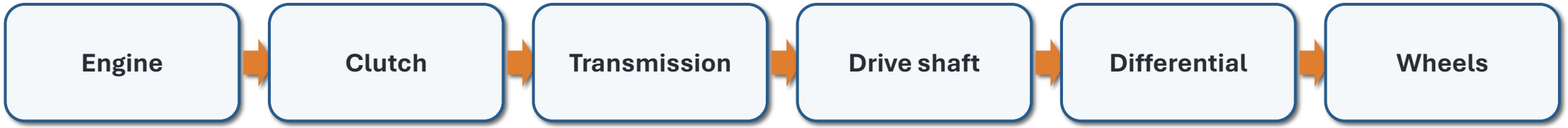


Figure 1.7 Transmission system.



Power Flow Through the Vehicle

The drive train indicates the power flow from the engine to the driving wheels.



Power source

Torque / speed conversion

Torque distribution

ISO-Based Vehicle Classification & Drive Layouts

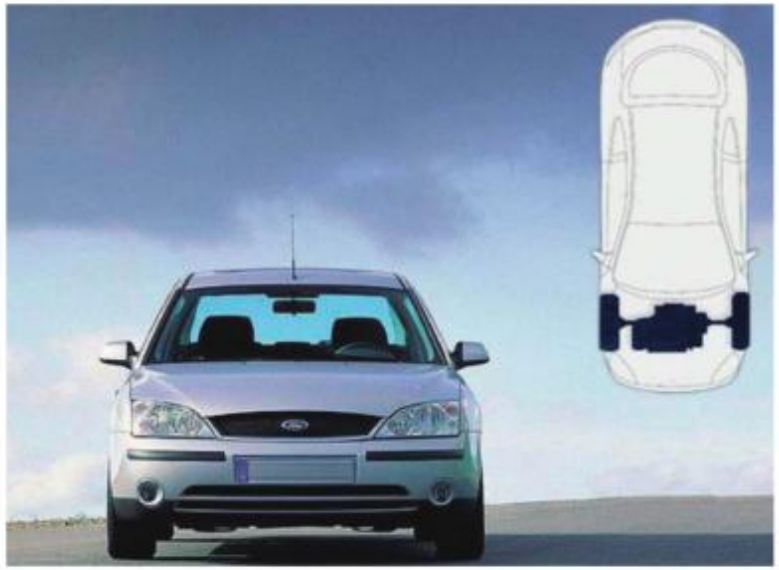


Figure 1.1 Front-engine, front-wheel drive

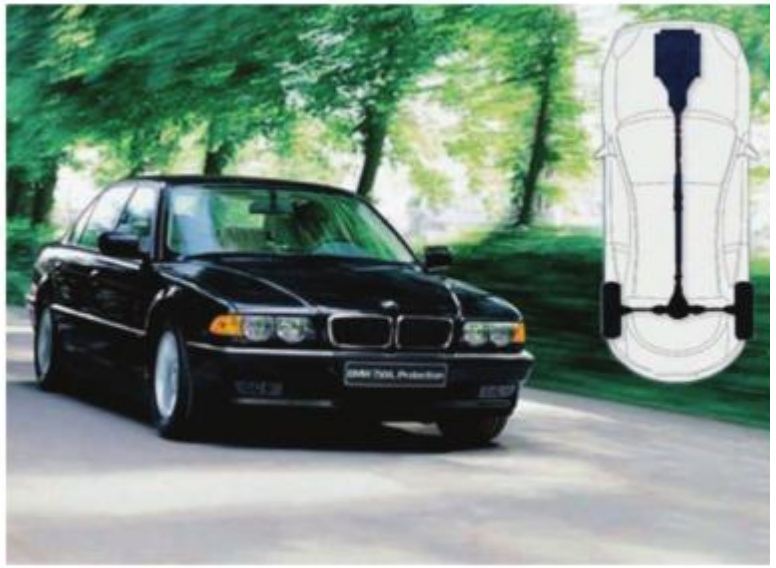


Figure 1.2 Front-engine, rear-wheel drive



Figure 1.3 Rear-engine, rear-wheel drive



Figure 1.4 Mid-engine, rear-wheel drive

Transmission system

In automobiles the rotary power (or torque) is obtained from the engine. The developed power is transmitted to the rotating wheels on the road. The unit which has various components connected in between the engine and the rear wheels of the vehicle is called as the automobile power transmission system. Let us see the important parts of an automobile transmission system.

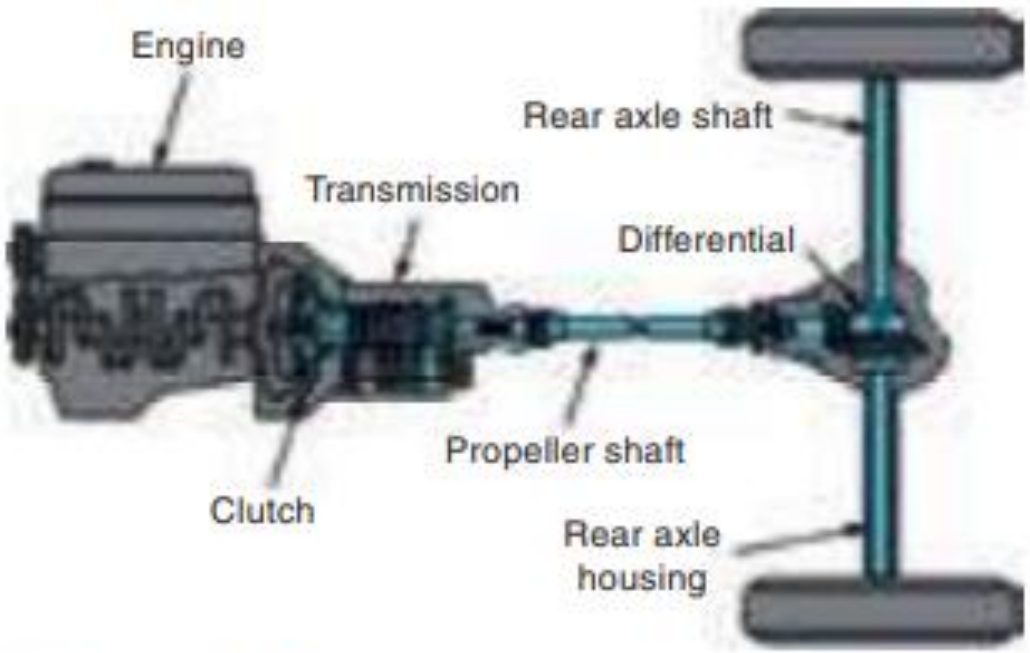


Figure 1.1.1 Important components of a transmission system of an automobile

Parts of Transmission System

1. Clutch
2. Gear box
3. Propeller shaft
4. Universal joint
5. Differential
6. Rear axle

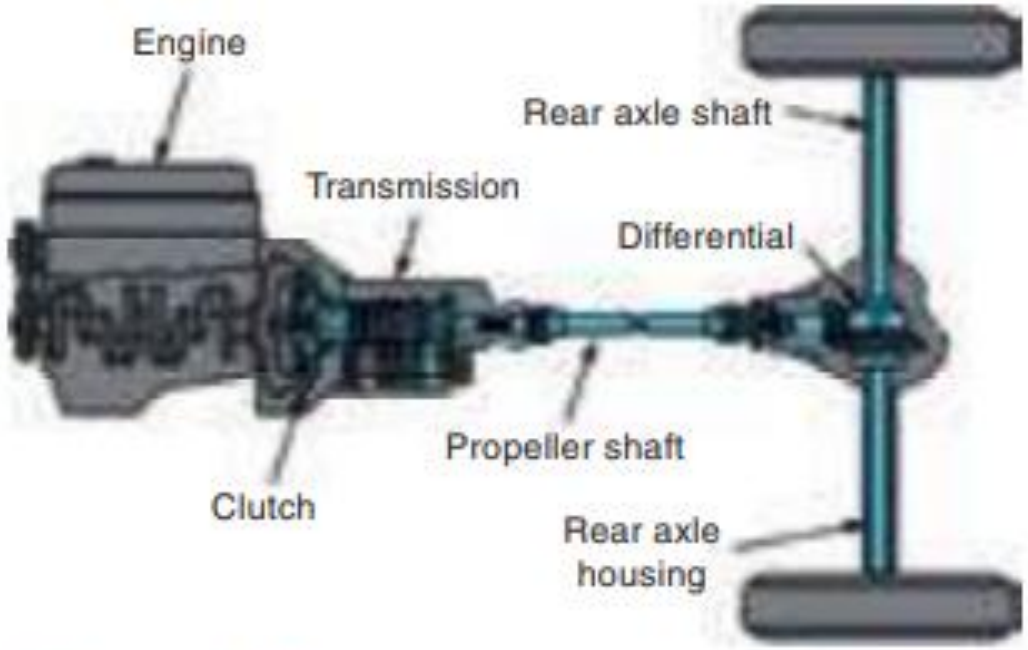


Figure 1.1.1 Important components of a transmission system of an automobile

Needs of transmission system

- For transmitting and stopping torque developed from the engine to the rear axle of the vehicle.
- For transmitting power to wheels without any noise during the engine is running.
- For avoiding vibrational effects during the transmission of power from engine to the rear wheels.
- For operating the vehicle wheels at the required speed.
- For balancing the speed of the wheels during vehicle is in running.

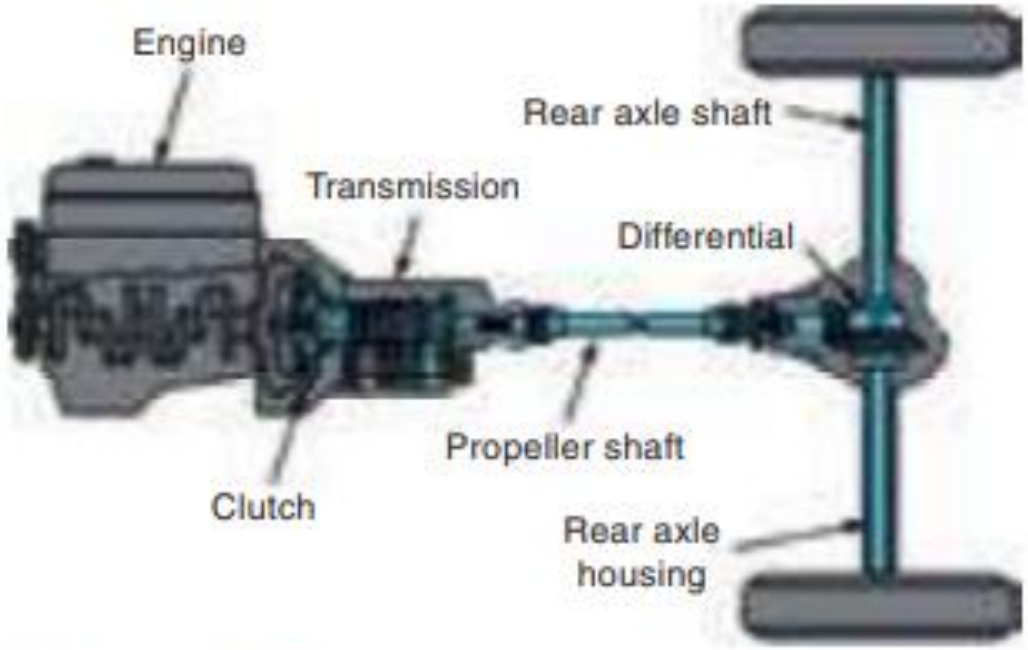


Figure 1.1.1 Important components of a transmission system of an automobile

Types of transmission system

1. Front Engine Rear Wheel Drive
2. Front Engine Front Wheel Drive
3. Front Engine Four Wheel Drive
4. Rear Engine Rear Wheel Drive

Front engine rear wheel drive

Advantages

- Engine cooling is good
- Vehicle's weight is evenly distributed
- It is easy to connect the components with the engine
- It is easy to control the vehicle

Disadvantages

- Lengthy propeller shaft is needed
- Vehicle's ground clearance is high
- Vehicle's stability is low.

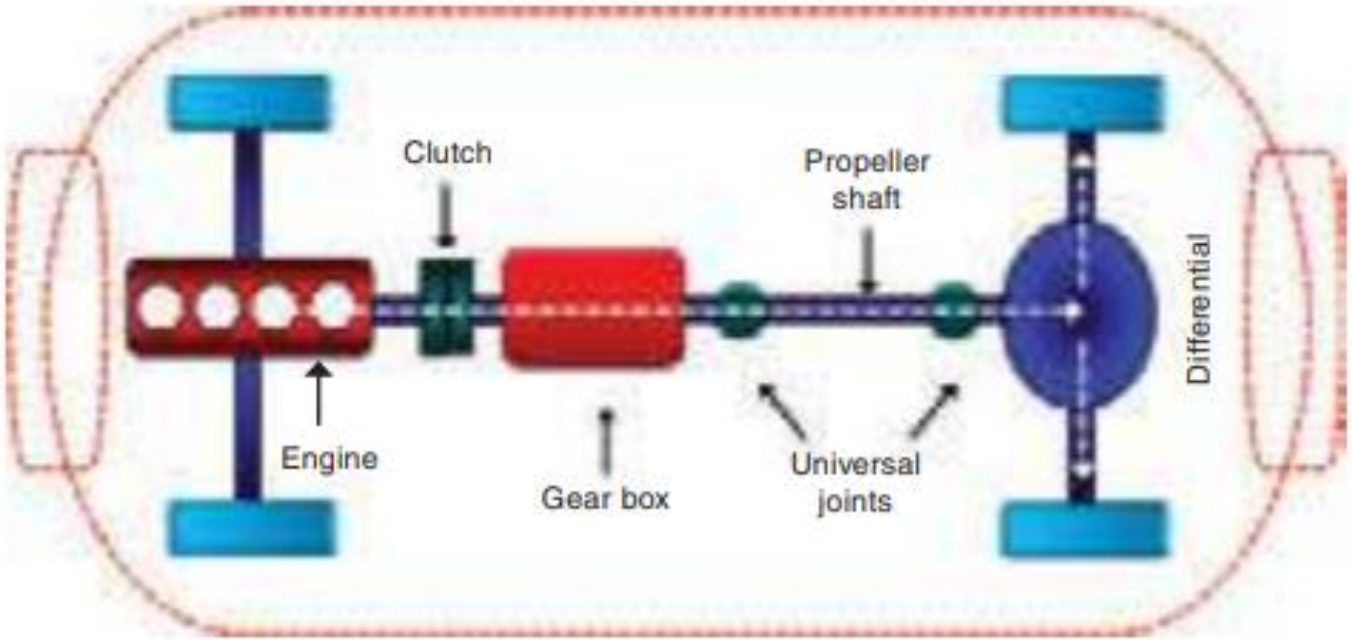


Figure 1.3.1 Front engine rear wheel drive

Front engine front wheel drive

Advantages

- Simple in design
- Have high stability
- Good road grip
- High engine cooling
- Good control over the vehicle
- There is no need of a separate propeller shaft

Disadvantages

- Maintenance cost is high
- Road grip is less when the vehicle is driven in slope or slanting regions
- Due to the engine's exhaust, passengers gets affected

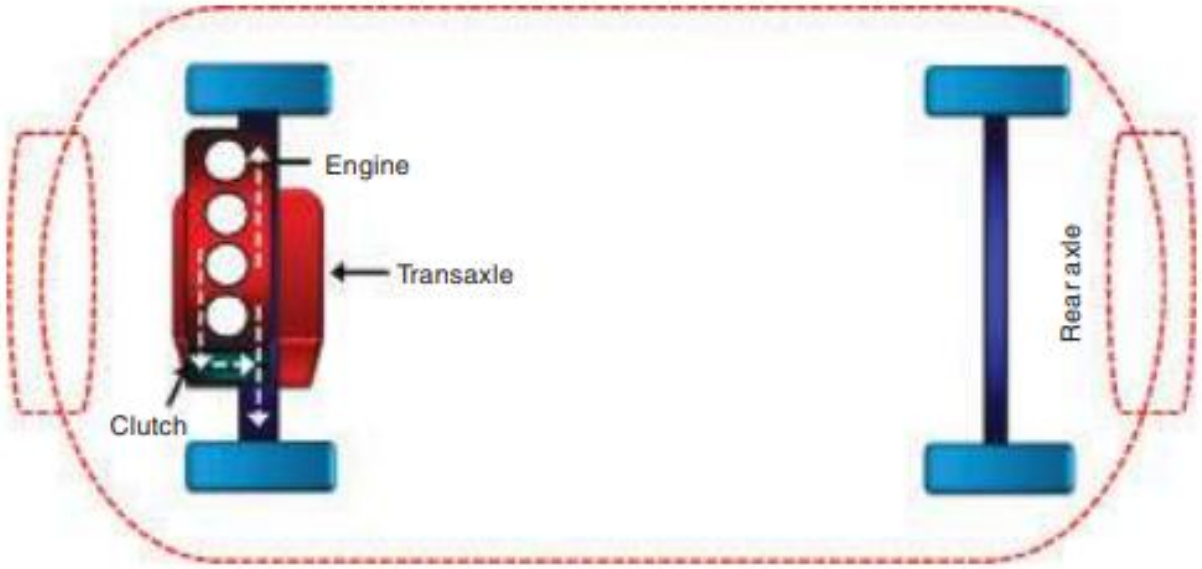


Figure 1.3.2 Front engine front wheel drive

Front engine four wheel drive

Advantages

- Pulling power is more
- Can operate vehicles in water, mud, clay and slime areas also
- Sometimes if one wheel fails to function, vehicles can operate using other three wheels

Disadvantages

- Initial cost is high
- Maintenance is high
- Complicated design
- Create more noise
- Vibration is more

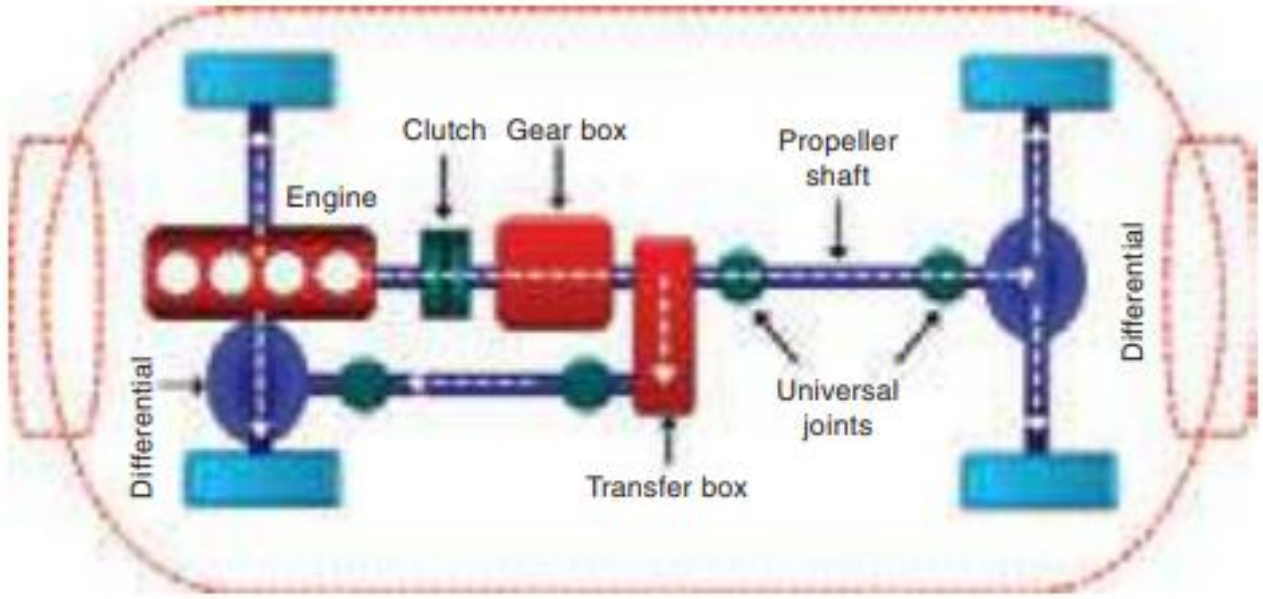


Figure 1.3.3 (a) Front engine four wheel drive

Rear engine rear wheel drive

Advantages

- Simple in design
- Separate propeller shaft is not needed
- Brake power is high
- No problem to passengers because of exhaust
- No need for lengthy propeller shaft

Disadvantages

- Stability is less
- Long linkages are needed
- Maintenance is high
- Engine cooling is less.

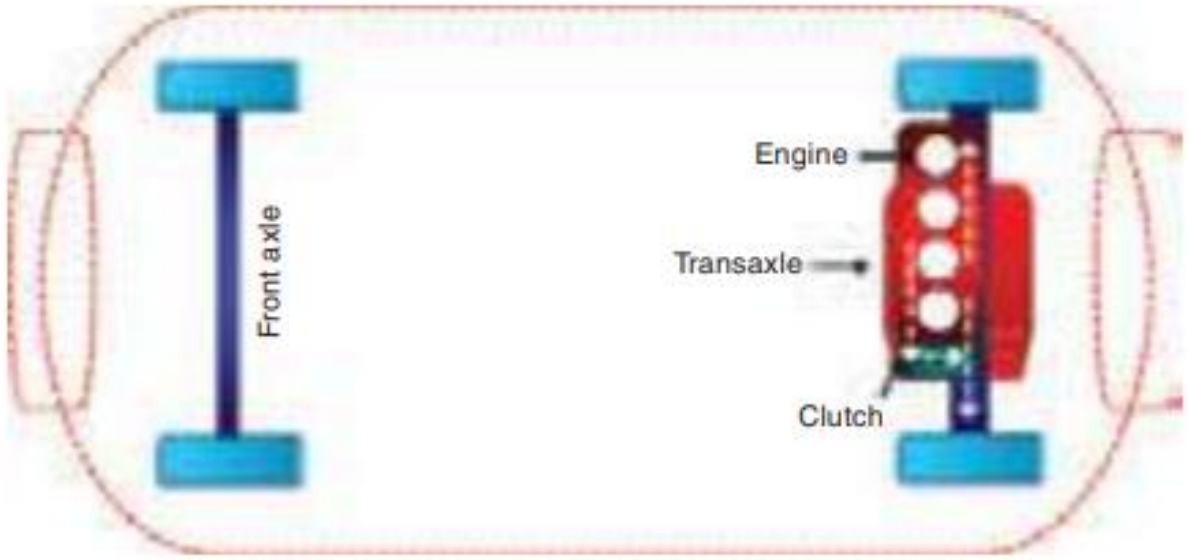
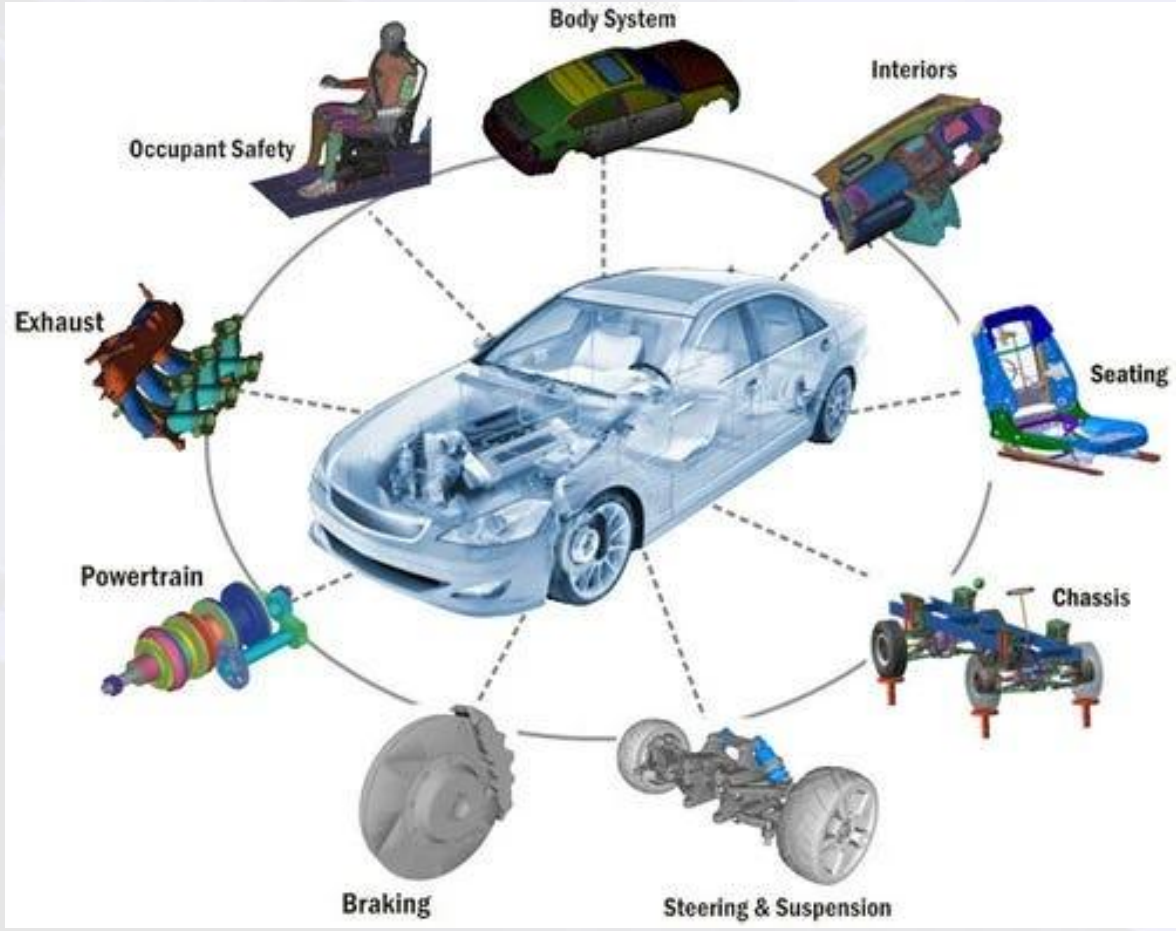


Figure 1.3.4 Rear engine Rear wheel drive

Basic components of Automobile

1. Frame
2. Engine
3. Clutch
4. Gear box
5. Propeller shaft
6. Rear axle
7. Front axle
8. Suspension system
9. Steering system
10. Braking system
11. Wheel and tyre and
12. The electrical system



Frame

The vehicle without the body is called as the Chassis. It is the under part of an automobile on which the body is mounted. The Chassis is called as the back bone of the vehicle. The structure of the chassis is shown in the above Figure 1.8.1.

It is a lengthy component which has cross frames joined rigidly by rivets. It is the foundation for carrying the engine, transmission system, steering system. This frame is made of stainless steel which is able to carry the whole weight of the vehicle. It is back bone of the vehicle.

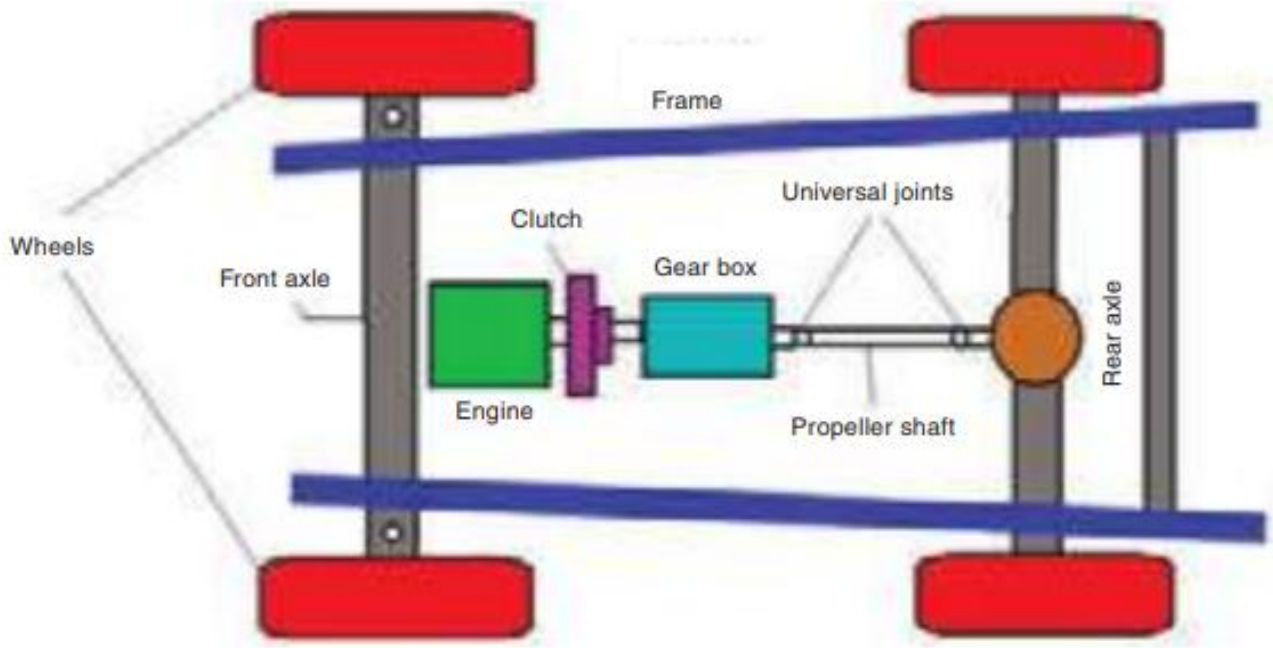


Figure 1.8.1 Important components of the Chassis

Engine

The necessary power can be produced from the engine for running the vehicle. Petrol and Diesel are used as fuels to run the engine. It is located at the front of the vehicle. It converts the heat energy by burning the fuel into mechanical energy. Hence it is called as the heart of the vehicle.



Figure 1.8.3 Engine

Clutch

Clutch is a mechanical device which is used to transfer and cut the engine's power transmitted to the driveline. It is located in between engine and gear box.



Figure 1.8.4 Clutch

Gear box

The box with set of gears inside the casing is called as gear box. It is used to change (increase and reduce) the speed and torque of vehicle according to different road and load conditions of the vehicle. It is shown Figure 1.8.5.



Figure 1.8.5 Gear box

Propeller Shaft

It is a type of long hollow shaft. This is located in between the gear box and differential. It transmits power from gear box to differential. It is shown Figure 1.8.6.



Figure 1.8.6 Propeller shaft

Front axle

Front wheels are coupled with front axle. Front axle carries the weight of the front part of the vehicle as well as facilitates steering and absorbs shocks due to road surface variations. It is shown Figure 1.8.7.



Figure 1.8.7 Front axle

Rear axle and Differential

Rear wheels are attached to the two ends of rear axle. The rear axle helps to turn the vehicle into the curved road. The differential gear in the center is cut by the speed of the cycle through the propeller shaft and then transported to 90° by wheels. It is shown Figure 1.8.8.



Figure 1.8.8 Rear Axle and Differential

Suspension system

The Spring Shock absorber has been connected to the front and back axles and helps to drive the vehicle smooth and slower from the vibrations in the road. It is shown Figure 1.8.9.



Figure 1.8.9 Suspension system

Steering system

The steering system operates on a very simple mechanism that directly transforms steering wheel rotation into straight line movement and helps in turn the vehicle as per the road. It is shown Figure 1.8.10.



Figure 1.8.10 Steering system

Braking system

Brakes are installed on the four wheels of the vehicle and work through mechanical connections, hydraulic and air. The very important part of an automobile is the braking system that helps to drive and control the speed of the vehicle within safe speed. It is shown Figure 1.8.11.

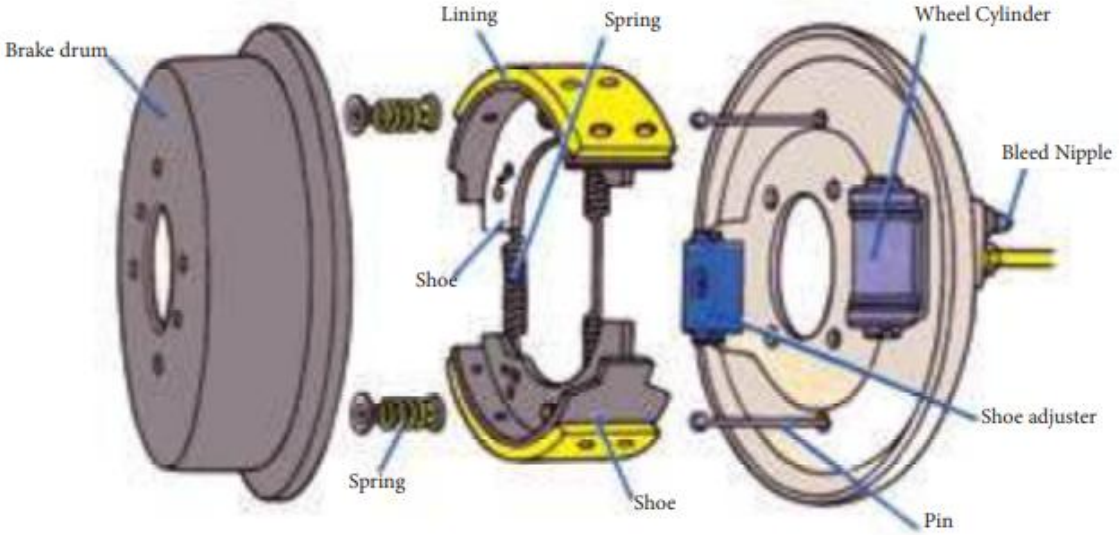


Figure 1.8.11 Braking system

Wheels and tyres

The wheels help to carry the vehicle's entire weight and to drive the vehicle. The wheels are connected to the front and rear axles of a vehicle. They absorb the vibrations produced from



Electrical system

Electrical system supplies electrical energy which is used to produce the spark for igniting the air fuel mixture in petrol engines, to provide light for the vehicle during night ride, to operate the wiper motor, to light inside the vehicle, to play music and for dashboard lights. It is shown Figure 1.8.13.

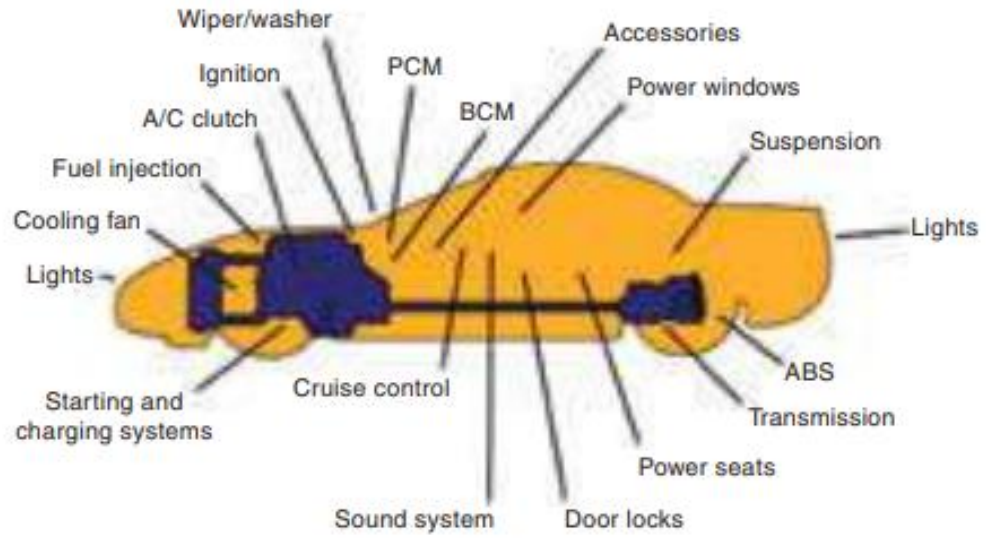


Figure 1.8.13 Basic overview of the electrical system of a modern car



Thank You!